

Multiblock approach for the passive scalar thermal lattice Boltzmann method

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A multiblock approach for the passive scalar thermal lattice Boltzmann method (TLBM) with multiple-relaxation-time collision scheme is proposed based on the Chapman-Enskog analysis. The interaction between blocks is executed in the moment space directly and an external force term is considered. Theoretical analysis shows that all the nonequilibrium parts of the nonconserved moments should be rescaled, while the nonequilibrium parts of the conserved moments can be calculated directly. Moreover, a local scheme based on the pseudoparticles for computing heat flux is proposed with no need to calculate temperature gradient based on the finite-difference scheme. In order to validate the multiblock approach and local scheme for computing heat flux, thermal Couette flow with wall injection is simulated and good results are obtained, which show that the adoption of the multiblock approach does not deteriorate the convergence rate of TLBM and the local scheme for computing heat flux has second-order convergence rate. Further application of the present approach is the simulation of natural convection in a square cavity with the Rayleigh number up to 10^9 .

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I. INTRODUCTION

As a powerful numerical technique for simulating fluid flows and complex physical processes in fluids, the lattice Boltzmann equation (LBE) method [1–3] has attracted considerable attention in past years. Different from the traditional computational fluid dynamics (CFD) method based on the macroscopic continuum equations, the LBE method can be regarded as a mesoscopic method based on the pseudoparticles, although it has been proved that LBE is a special discrete form of the Boltzmann equation (BE) [4,5]. As an extremely important problem in engineering and natural systems, thermal fluid flow has also been solved by the LBE method extensively [6–15].

The existing thermal lattice Boltzmann methods (TLBMs) can generally be grouped into three major categories: (1) the multispeed approach [6,7], (2) the double-distribution-function (DDF) approach [8–13], and (3) the hybrid approach [14,15]. The multispeed approach is a straightforward extension of the athermal LBE method, and uses only one distribution function for the thermodynamics equations. The DDF approach uses two independent distribution functions: one for the velocity field and one for the temperature field. The hybrid approach is a mixed method, where the velocity field is solved by the LBE method, and the temperature field is solved by conventional numerical techniques (such as the finite-difference method, etc.). Among these TLBMs, the DDF approach is employed most commonly due to its good numerical stability, adjustable Prandtl number, and easy implementation.

The DDF approaches can also be further classified into two main categories: (1) the passive scalar approach [8–10] and (2) the thermal energy distribution approach [11–13]. In the passive scalar approach, the viscous dissipation and compression work are commonly ignored because this

approach is constructed artificially merely to recover the correct macroscopic equation [i.e., the convection-diffusion equation (CDE)]. Whereas in the thermal energy distribution approach, both the viscous dissipation and compression work can be considered naturally because this approach is derived from the Boltzmann equation for the internal energy or total energy directly. However, the thermal energy distribution approach is much more complicated than the passive scalar approach. And as Peng *et al.* [10] pointed out, the viscous dissipation and compression work can be neglected in real incompressible applications. Therefore the passive scalar approach is more widely adopted in real applications due to its simplicity [16–18].

In order to solve thermal fluid flow more efficiently by the passive scalar TLBM, the multiblock approach with grid refinement can be used. Unlike the traditional CFD methods, there exist a relatively small number of papers treating grid refinement in the framework of the LBE method [19–25]. In 1998, Filippova and Hänel [21] first introduced the grid refinement technique in the LBE method. A two-way interaction approach with rescaling the postcollision distribution functions was proposed. Later, Yu *et al.* [22] developed Filippova and Hänel's method further, adopting the cubic spline fitting for spatial interpolation and the three-point Lagrangian formula for temporal interpolation. Different from Filippova and Hänel's method, Lin and Lai [23] proposed a simplified method with a one-way interaction approach and assumed that the distribution functions are interchangeable between different grids. Having pointed out that there exists a singularity ($\tau = 1$) in Filippova and Hänel's method, and that Lin and Lai's method is inaccurate since the nonequilibrium parts of the distribution functions are not interchangeable, Dupuis and Chopard [24] proposed a more general two-way interaction approach with rescaling the poststreaming distribution functions. In 2012, Lagrava *et al.* [25] found that a direct copy of the information cannot guarantee the numerical stability when the Reynolds number is high, and then proposed a filter technique to remove the scales of the fine grid that are not resolved by the coarse grid.

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It should be pointed out that most of the grid refinement approaches in the LBE method are proposed for athermal fluid flows without considering an external force term in LBEs [21–25]. What is more, a single-relaxation-time (SRT) collision scheme is adopted in the above LBEs. Although some scholars have applied the grid refinement approach to LBEs with a multiple-relaxation-time (MRT) collision scheme [26,27], they only deduced the interaction approach based on the continuity of mass, momentum, and stresses between different grids. In this paper, a multiblock approach with grid refinement for the passive scalar TLBM with MRT collision scheme is proposed. The interaction approach with considering an external force term is deduced based on the Chapman-Enskog analysis in the moment space directly. Moreover, as an important physical variable, heat flux has not obtained much attention in the framework of the LBE method. Here, a local scheme (“local” means the computation

process at one point only depends on the information of this point) based on the pseudoparticles is proposed for computing heat flux with no need to calculate temperature gradient based on the finite-difference scheme. The remainder of the present paper is organized as follows. Section II presents the passive scalar TLBM with MRT collision scheme, and then the corresponding multiblock approach is given in Sec. III. Section IV introduces a local scheme for computing heat flux. Some numerical tests are carried out in Sec. V. Finally, a brief conclusion is drawn in Sec. VI.

II. THERMAL LATTICE BOLTZMANN METHOD

Without loss of generality, the two-dimensional nine-velocity (D2Q9) model is adopted for both the velocity and temperature fields. The nine discrete velocities are given by

$$\mathbf{e}_i = \begin{cases} c(0,0), & i = 0, \\ c(\cos[(i-1)\pi/2], \sin[(i-1)\pi/2]), & i = 1,2,3,4, \\ \sqrt{2}c(\cos[(2i-1)\pi/4], \sin[(2i-1)\pi/4]), & i = 5,6,7,8, \end{cases} \quad (1)$$

where $c = \delta_x/\delta_t$ is the lattice speed, in which δ_x is the lattice spacing, and δ_t is the time step. Note that δ_x and δ_t are dimensionless, and $c \equiv 1$ is appointed in order to simplify the linear transformation processes between the velocity and moment spaces. In what follows all variables are defined to be dimensionless too. The LBE with MRT collision scheme for the density and temperature distribution functions can be expressed as

$$f_i^\sigma(\mathbf{x} + \mathbf{e}_i\delta_t, t + \delta_t) = f_i^\sigma(\mathbf{x}, t) - \Lambda_{ij}^\sigma [f_j^\sigma(\mathbf{x}, t) - f_j^{\sigma, \text{eq}}(\mathbf{x}, t)] + \delta_t F_{vi}^\sigma(\mathbf{x}, t), \quad i = 0-8, \quad (2)$$

which can be decomposed into two substeps, i.e., the collision process

$$\tilde{f}_i^\sigma(\mathbf{x}, t) = f_i^\sigma(\mathbf{x}, t) - \Lambda_{ij}^\sigma [f_j^\sigma(\mathbf{x}, t) - f_j^{\sigma, \text{eq}}(\mathbf{x}, t)] + \delta_t F_{vi}^\sigma(\mathbf{x}, t), \quad i = 0-8 \quad (3)$$

and the streaming process

$$f_i^\sigma(\mathbf{x} + \mathbf{e}_i\delta_t, t + \delta_t) = \tilde{f}_i^\sigma(\mathbf{x}, t), \quad i = 0-8. \quad (4)$$

Here, f_i^σ is the distribution function (i.e., the poststreaming distribution function), $\tilde{f}_i^\sigma$ is the postcollision distribution function, $f_i^{\sigma, \text{eq}}$ is the equilibrium distribution function, and Λ^σ and $\mathbf{F}_v^\sigma$ are the relaxation matrix and discrete force term in the velocity space, respectively. The superscript “ σ ” denotes the velocity field when $\sigma = u$ or the temperature field when $\sigma = T$. Generally, the collision process is carried out in the moment space as [28–30]

$$\tilde{\mathbf{m}}^\sigma(\mathbf{x}, t) = \mathbf{m}^\sigma(\mathbf{x}, t) - \mathbf{S}^\sigma [\mathbf{m}^\sigma(\mathbf{x}, t) - \mathbf{m}^{\sigma, \text{eq}}(\mathbf{x}, t)] + \delta_t \left(\mathbf{I} - \frac{\mathbf{S}^\sigma}{2} \right) \mathbf{F}_m^\sigma, \quad (5)$$

where $\mathbf{I}$ is the unit matrix, $\mathbf{m}^\sigma = \mathbf{M} \mathbf{f}^\sigma$ is the moment calculated by the distribution function through a linear transformation by the orthogonal transformation matrix $\mathbf{M}$ [28], $\tilde{\mathbf{m}}^\sigma = \mathbf{M} \tilde{\mathbf{f}}^\sigma$ is

the postcollision moment, $\mathbf{m}^{\sigma, \text{eq}}$ is the equilibrium moment function, and $\mathbf{S}^\sigma = \mathbf{M} \Lambda^\sigma \mathbf{M}^{-1}$ and $\mathbf{F}_m^\sigma = (\mathbf{I} - \mathbf{S}^\sigma/2)^{-1} \mathbf{M} \mathbf{F}_v^\sigma$ are the relaxation matrix and discrete force term in the moment space, respectively. In general, the relaxation matrix in the moment space is a diagonal matrix, which can be expressed as

$$\mathbf{S}^\sigma = \text{diag}(s_0^\sigma, s_e^\sigma, s_\epsilon^\sigma, s_j^\sigma, s_q^\sigma, s_j^\sigma, s_q^\sigma, s_p^\sigma, s_p^\sigma), \quad (6)$$

and the equilibrium moment function can be given as [28]

$$\mathbf{m}^{u, \text{eq}} = (\rho, -2\rho + 3u^2, \rho - 3u^2, u_x, -u_x, u_y, -u_y, u_x^2 - u_y^2, u_x u_y)^T, \quad (7a)$$

$$\mathbf{m}^{T, \text{eq}} = (T, -2T, T, T u_x, -T u_x, T u_y, -T u_y, 0, 0)^T, \quad (7b)$$

where the superscript “ T ” means the transposition of a vector. Note that here an incompressible LBE method with modified equilibrium moment function is adopted, and the mean density (ρ_0) is set to be 1.0 [28]. In order to avoid error terms induced by the unsteady and nonuniform force, the discrete force term in the moment space should be given as [29,30]

$$\mathbf{F}_m^u = (0, 6\mathbf{F} \cdot \mathbf{u}, -6\mathbf{F} \cdot \mathbf{u}, F_x, -F_x, F_y, -F_y, 2F_x u_x - 2F_y u_y, F_x u_y + F_y u_x)^T, \quad (8a)$$

$$\mathbf{F}_m^T = (q, -2q, q, 0, 0, 0, 0, 0, 0)^T, \quad (8b)$$

where $\mathbf{F} = (F_x, F_y)^T$ is the force term in the momentum equation and q is the heat source term in the energy equation. With the assumption of the Boussinesq approximation, $\mathbf{F}$ can be expressed as

$$\mathbf{F} = -\mathbf{g}\beta(T - T_{\text{ref}}), \quad (9)$$

in which $\mathbf{g}$ is the acceleration of gravity, β is the volume expansivity, and T_{ref} is the reference temperature. The corresponding macroscopic variables such as density, velocity, and

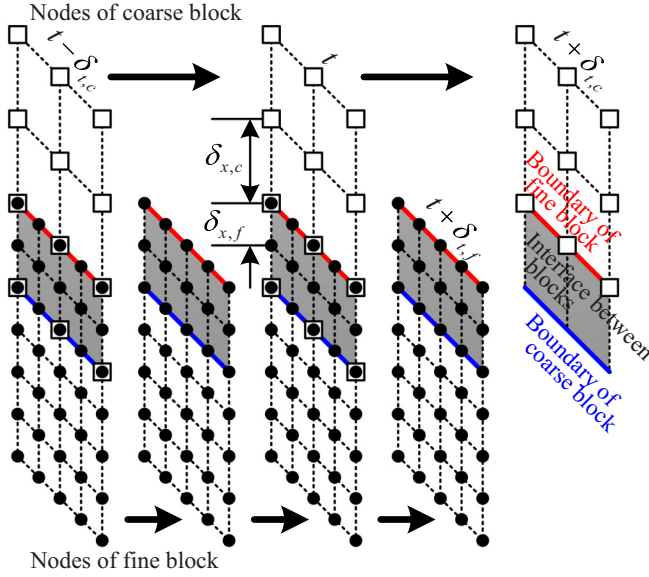


FIG. 1. (Color online) Interface structure between two blocks of different lattice spacings.

temperature are calculated via

$$\rho = \sum_{i=0}^8 f_i^u, \quad \mathbf{u} = \sum_{i=0}^8 \mathbf{e}_i f_i^u + \frac{\delta_t}{2} \mathbf{F}, \quad T = \sum_{i=0}^8 f_i^T + \frac{\delta_t}{2} q. \quad (10)$$

Through the Chapman-Enskog analysis, the corresponding macroscopic continuum equations are

$$\partial_t \rho + \nabla \cdot \mathbf{u} = 0, \quad (11a)$$

$$\partial_t \mathbf{u} + \nabla \cdot (\mathbf{u}\mathbf{u}) = -\nabla p + \nu \nabla^2 \mathbf{u} + \zeta \nabla (\nabla \cdot \mathbf{u}) + \mathbf{F}, \quad (11b)$$

$$\partial_t T + \nabla \cdot (T\mathbf{u}) = \alpha \nabla^2 T + q, \quad (11c)$$

where the pressure $p = \rho/3$, the kinetic viscosity $\nu = \delta_t(1/s_p^u - 0.5)/3$, the bulk viscosity $\zeta = \delta_t(1/s_e^u - 0.5)/3$,

and the thermal diffusivity $\alpha = \delta_t(1/s_f^T - 0.5)/3$. Note that $\rho_0 = 1.0$ has been substituted into Eqs. (11a) and (11b).

III. MULTIBLOCK APPROACH FOR MRT SCHEME

As illustrated by Fig. 1, a two-block system (i.e., the coarse and fine blocks) is considered here. The ratio of the lattice spacing and time step between the coarse and fine blocks are

$$n = \frac{\delta_{x,c}}{\delta_{x,f}} = \frac{\delta_{t,c}}{\delta_{t,f}}, \quad (12)$$

which indicates $c \equiv 1$ for both blocks. Here, the subscript “c” denotes the coarse block and subscript “f” denotes the fine block. Note that the overlap area (the gray area in Fig. 1) is set for the convenience of information exchange (i.e., interaction approach) between adjacent blocks, i.e., all information on the boundary of the coarse block (the blue solid line in Fig. 1) can be obtained from the fine block, and similarly, all information on the boundary of the fine block (the red solid line in Fig. 1) can be obtained from the coarse block.

To deduce the interaction approach for the MRT collision scheme, the Chapman-Enskog analysis is conducted here. Applying the Taylor series expansion to Eq. (2) and rewriting it in the moment space, we can get

$$\begin{aligned} \tilde{\mathbf{D}} \mathbf{m}^\sigma + \frac{\delta_t}{2} \tilde{\mathbf{D}}^2 \mathbf{m}^\sigma + O(\delta_t^2) \\ = -\frac{\mathbf{S}^\sigma}{\delta_t} (\mathbf{m}^\sigma - \mathbf{m}^{\sigma, \text{eq}}) + \left(\mathbf{I} - \frac{\mathbf{S}^\sigma}{2} \right) \mathbf{F}_m^\sigma, \end{aligned} \quad (13)$$

where $\tilde{\mathbf{D}} = \mathbf{I} \partial_t + \mathbf{E}_x \partial_x + \mathbf{E}_y \partial_y$, in which $\mathbf{E}_x = \mathbf{M}[\text{diag}(e_{0x}, \dots, e_{8x})] \mathbf{M}^{-1}$ and $\mathbf{E}_y = \mathbf{M}[\text{diag}(e_{0y}, \dots, e_{8y})] \mathbf{M}^{-1}$. Substituting the following Chapman-Enskog expansions

$$\partial_t = \varepsilon \partial_{t1} + \varepsilon^2 \partial_{t2}, \quad \partial_x = \varepsilon \partial_{x1}, \quad \partial_y = \varepsilon \partial_{y1}, \quad (14a)$$

$$\mathbf{m}^\sigma = \sum_{n=0}^{+\infty} \varepsilon^n \mathbf{m}^{\sigma, (n)}, \quad \mathbf{F}_m^\sigma = \varepsilon \mathbf{F}_m^{\sigma, (1)} \quad (14b)$$

into Eq. (13), we can rewrite it in the consecutive orders of the small expansion parameter ε as

$$\varepsilon^0: \mathbf{m}^{\sigma, (0)} = \mathbf{m}^{\sigma, \text{eq}}, \quad (15a)$$

$$\varepsilon^1: (\mathbf{I} \partial_{t1} + \mathbf{E}_x \partial_{x1} + \mathbf{E}_y \partial_{y1}) \mathbf{m}^{\sigma, (0)} - \mathbf{F}_m^{\sigma, (1)} = -\frac{\mathbf{S}^\sigma}{\delta_t} \left(\mathbf{m}^{\sigma, (1)} + \frac{\delta_t}{2} \mathbf{F}_m^{\sigma, (1)} \right), \quad (15b)$$

$$\varepsilon^2: \partial_{t2} \mathbf{m}^{\sigma, (0)} + (\mathbf{I} \partial_{t1} + \mathbf{E}_x \partial_{x1} + \mathbf{E}_y \partial_{y1}) \left(\mathbf{I} - \frac{\mathbf{S}^\sigma}{2} \right) \left(\mathbf{m}^{\sigma, (1)} + \frac{\delta_t}{2} \mathbf{F}_m^{\sigma, (1)} \right) = -\frac{\mathbf{S}^\sigma}{\delta_t} \mathbf{m}^{\sigma, (2)}. \quad (15c)$$

From Eq. (15), the macroscopic continuum equation [i.e., Eq. (11)] can be recovered. Therefore, in order to maintain the continuity of macroscopic properties across the block interface, Eq. (15) should be continuous between blocks fundamentally.

In the interaction approach between blocks, the moment can be decomposed into two parts: the equilibrium and nonequilibrium parts, that is,

$$\mathbf{m}^\sigma = \mathbf{m}^{\sigma, \text{eq}} + \mathbf{m}^{\sigma, \text{neq}}. \quad (16)$$

For the equilibrium part, since the macroscopic variables are independent of grids, Eqs. (7) and (8) indicate

$$\mathbf{m}_c^{\sigma, \text{eq}} = \mathbf{m}_f^{\sigma, \text{eq}}, \quad \mathbf{F}_{m,c}^\sigma = \mathbf{F}_{m,f}^\sigma. \quad (17)$$

As for the nonequilibrium part, the truncation $\mathbf{m}^{\sigma, (n)} \approx \mathbf{0}$ ($\forall n \geq 2$) should be applied, which indicates that the nonequilibrium part can be simplified as $\mathbf{m}^{\sigma, \text{neq}} \approx \varepsilon \mathbf{m}^{\sigma, (1)}$. Further, Eq. (17) is equivalent to $\mathbf{m}_c^{\sigma, (0)} = \mathbf{m}_f^{\sigma, (0)}$ and $\mathbf{F}_{m,c}^\sigma = \mathbf{F}_{m,f}^\sigma$ due to Eq. (15a) and $\mathbf{F}_m^\sigma = \varepsilon \mathbf{F}_m^{\sigma, (1)}$, and $c \equiv 1$ indicates that $\mathbf{E}_x$ and $\mathbf{E}_y$ remain consistent between blocks. Therefore,

the left-hand side of Eq. (15b) is independent of grids, which suggests the right-hand side of Eq. (15b) should satisfy

$$\frac{\mathbf{S}_c^\sigma}{\delta_{t,c}} \left(\mathbf{m}_c^{\sigma,(1)} + \frac{\delta_{t,c}}{2} \mathbf{F}_{m,c}^{\sigma,(1)} \right) = \frac{\mathbf{S}_f^\sigma}{\delta_{t,f}} \left(\mathbf{m}_f^{\sigma,(1)} + \frac{\delta_{t,f}}{2} \mathbf{F}_{m,f}^{\sigma,(1)} \right), \quad (18)$$

which can be further written as

$$\frac{\mathbf{S}_c^\sigma}{\delta_{t,c}} \left(\mathbf{m}_c^{\sigma,\text{neq}} + \frac{\delta_{t,c}}{2} \mathbf{F}_{m,c}^{\sigma} \right) = \frac{\mathbf{S}_f^\sigma}{\delta_{t,f}} \left(\mathbf{m}_f^{\sigma,\text{neq}} + \frac{\delta_{t,f}}{2} \mathbf{F}_{m,f}^{\sigma} \right). \quad (19)$$

It can be seen that the nonequilibrium parts should be rescaled between the coarse and fine blocks. Moreover, replacing the part $(\mathbf{m}_c^{\sigma,(1)} + \delta_{t,c} \mathbf{F}_{m,c}^{\sigma,(1)})/2$ in Eq. (15c) with Eq. (15b), and considering $\mathbf{m}_c^{\sigma,(n)} \approx \mathbf{0}$ ($\forall n \geq 2$), all the other parts of Eq. (15c) are independent of grids except $(\mathbf{I} - \mathbf{S}^\sigma/2) \mathbf{S}^{\sigma,-1} \delta_t$, which indicates that

$$\delta_{t,c} (\mathbf{S}_c^{\sigma,-1} - \mathbf{I}/2) = \delta_{t,f} (\mathbf{S}_f^{\sigma,-1} - \mathbf{I}/2). \quad (20)$$

Equations (17), (19), and (20) describe the interaction approach between the coarse and fine blocks in the moment space. Obviously, it can be seen from Eq. (19) that the external force term in LBEs should be considered in the interaction approach, while most previous works are proposed for LBEs without an external force term.

More attention should be focused on the conserved moments ($i = 0, 3, 5$ for $\sigma = u$ and $i = 0$ for $\sigma = T$, i.e., m_0^u, m_3^u, m_5^u , and m_0^T). For these conserved moments, substituting the Chapman-Enskog expansions into Eq. (10), we can get

$$m_i^{\sigma,(1)} + \frac{\delta_t}{2} F_{m,i}^{\sigma,(1)} \equiv 0, \quad m_i^{\sigma,(n)} \equiv 0 \quad (\forall n \geq 2), \quad (21)$$

which means that $m_i^{\sigma,\text{neq}} = \varepsilon m_i^{\sigma,(1)}$ and $m_i^{\sigma,\text{neq}} + \delta_t F_{m,i}^{\sigma}/2 \equiv 0$, namely, the nonequilibrium parts of the conserved moments can be calculated precisely and directly and there is no need to execute the rescaling procedure [Eq. (19)] for them. Further, in view of Eq. (21), all terms containing the corresponding relaxation factors (i.e., s_0^u, s_j^u , and s_0^T) in Eq. (15c) disappear, which indicates there is no need to rescale these relaxation factors as Eq. (20) describes, namely, we can set $s_0^u = s_j^u = s_0^T \equiv 1$ for both blocks consistently. From the above discussions, we can see that the rescaling procedures [Eqs. (19) and (20)] are only valid for the nonconserved moments. Note that in the interaction approach for the MRT collision scheme proposed previously [26,27], only the nonequilibrium parts of the moments m_7^u and m_8^u are rescaled for the continuity of stresses; while in the present approach, all the nonequilibrium parts of the nonconserved moments are rescaled from the perspective of the continuity of the first-order expansion equation [i.e., Eq. (15b)], which may be more fundamental.

In real simulation, the interaction approach can be carried out by transferring the information from the coarse block to the fine block, or alternatively from the fine block to the coarse block in the following sequence: (1) transfer the macroscopic variables ($\rho, \mathbf{u}, T, \mathbf{F}$, and q) which are independent of grids; (2) transfer the equilibrium part of the moment ($\mathbf{m}^{\sigma,\text{eq}}$) and the discrete force term ($\mathbf{F}_m^\sigma$) based on Eq. (17); (3) transfer the

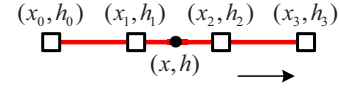


FIG. 2. (Color online) Schematic of spatial and temporal interpolations.

nonequilibrium part of the moment ($\mathbf{m}^{\sigma,\text{neq}}$) based on Eq. (19) in which the relation between the relaxation matrices is given by Eq. (20); and (4) calculate the moment by Eq. (16). As shown by Fig. 1, when we transfer information from the coarse block to the fine block, spatial and temporal interpolations are needed on the boundary of the fine block because both the lattice spacing and time step for the fine block are n times smaller than those of the coarse block. As pointed out by Lagrava *et al.* [25], the second-order interpolation cannot guarantee the mass conservation and third-order interpolation is needed at least. Accordingly, the n -point Lagrangian formula with n th-order accuracy is adopted here. For spatial interpolation, four points are used; for temporal interpolation, three points are used. As illustrated by Fig. 2, the n -point Lagrangian formula is expressed as

$$h(x) = \sum_{i=0}^{n-1} h_i \left(\prod_{\substack{j=0 \\ j \neq i}}^{n-1} \frac{x - x_j}{x_i - x_j} \right) \quad (n = 3, 4). \quad (22)$$

Note that, for temporal interpolation, (x_0, h_0) , (x_1, h_1) , (x_2, h_2) , and (x_3, h_3) denote the information obtained from the coarse block at times $t - \delta_{t,c}$, t , $t + \delta_{t,c}$, and $t + 2\delta_{t,c}$, respectively, and (x, h) denotes the information at time $t + k\delta_{t,f}$ ($0 < k < n$). Therefore, only (x_0, h_0) , (x_1, h_1) , and (x_2, h_2) are used for the temporal interpolation (see Fig. 1) and the three-point Lagrangian formula is adopted.

IV. LOCAL SCHEME FOR COMPUTING HEAT FLUX

From the point of collision and streaming processes, the temperature distribution function can be regarded as pseudoparticles which are composed of “thermal energy”, and the heat flux can be treated as particle flux (note that the “heat flux” in this paper actually means the “temperature flux”). So that the heat flux in the x direction at node $\mathbf{x}$ can be calculated by

$$J_{x,L}(\mathbf{x}, t + \delta_t) = [(f_1^T + f_5^T + f_8^T)|_{(\mathbf{x}, t + \delta_t)} - (\bar{f}_3^T + \bar{f}_6^T + \bar{f}_7^T)|_{(\mathbf{x}, t)}] \delta_x / \delta_t, \quad (23a)$$

$$J_{x,R}(\mathbf{x}, t + \delta_t) = [(\bar{f}_1^T + \bar{f}_5^T + \bar{f}_8^T)|_{(\mathbf{x}, t)} - (f_3^T + f_6^T + f_7^T)|_{(\mathbf{x}, t + \delta_t)}] \delta_x / \delta_t, \quad (23b)$$

$$J_x(\mathbf{x}, t + \delta_t) = \frac{J_{x,L}(\mathbf{x}, t + \delta_t) + J_{x,R}(\mathbf{x}, t + \delta_t)}{2}. \quad (23c)$$

Here, $J_{x,L}(\mathbf{x}, t + \delta_t)$ and $J_{x,R}(\mathbf{x}, t + \delta_t)$ are considered as the heat fluxes at the virtual points (i.e., points L and R shown

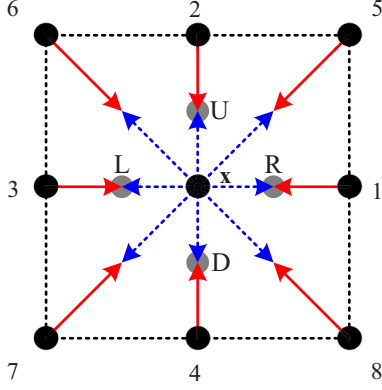


FIG. 3. (Color online) Streaming process at node $\mathbf{x}$ with the black solid circles being the lattice nodes, the gray solid circles being the virtual points, the blue dashed lines being the postcollision distribution functions $\bar{f}_i^T(\mathbf{x}, t)$, and the red solid lines being the poststreaming distribution functions $f_i^T(\mathbf{x}, t + \delta_t)$.

in Fig. 3) at the left and right sides of node $\mathbf{x}$, respectively. Similarly, the heat flux in the y direction at node $\mathbf{x}$ can be calculated by

$$J_{y,D}(\mathbf{x}, t + \delta_t) = [(\bar{f}_2^T + \bar{f}_5^T + \bar{f}_6^T)|_{(\mathbf{x}, t + \delta_t)} - (\bar{f}_4^T + \bar{f}_7^T + \bar{f}_8^T)|_{(\mathbf{x}, t)}] \delta_x / \delta_t, \quad (24a)$$

$$J_{y,U}(\mathbf{x}, t + \delta_t) = [(\bar{f}_2^T + \bar{f}_5^T + \bar{f}_6^T)|_{(\mathbf{x}, t)} - (\bar{f}_4^T + \bar{f}_7^T + \bar{f}_8^T)|_{(\mathbf{x}, t + \delta_t)}] \delta_x / \delta_t, \quad (24b)$$

$$J_y(\mathbf{x}, t + \delta_t) = \frac{J_{y,D}(\mathbf{x}, t + \delta_t) + J_{y,U}(\mathbf{x}, t + \delta_t)}{2}. \quad (24c)$$

Here, $J_{y,D}(\mathbf{x}, t + \delta_t)$ and $J_{y,U}(\mathbf{x}, t + \delta_t)$ are considered as the heat fluxes at the virtual points (i.e., points D and U shown in Fig. 3) at the lower and upper sides of node $\mathbf{x}$, respectively.

As seen from the above discussion, the present computation process for heat flux is local, and the calculation of temperature gradient based on the finite-difference scheme can be avoided. Compared with the very recent work proposed by Chai and Zhao based on the Chapman-Enskog analysis [31], an easy and concise alternative way is proposed here.

V. RESULTS AND DISCUSSIONS

In this section, numerical tests are conducted to validate the present multiblock approach for the TLBM with MRT collision scheme and local scheme for computing heat flux. The simulated problems include the thermal Couette flow with wall injection and the natural convection in a square cavity with the Rayleigh number up to 10^9 .

A. Thermal Couette flow with wall injection

In order to validate the present multiblock approach and local scheme for computing heat flux, a simulation of thermal Couette flow with wall injection (which has analytical solution) is carried out in this section. As shown by Fig. 4, the problem consists of a parallel channel with porous walls,

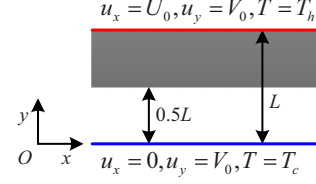


FIG. 4. (Color online) Schematic diagram of the multiblock arrangement for the thermal Couette flow with boundary conditions shown (the gray area is the fine block with $n = 4$, and the white area is the coarse block).

and the fluid is injected into the channel from the stationary bottom wall with velocity V_0 , and withdrawn from the moving top wall with the same velocity. The problem has the following analytical solutions for the velocity and temperature as [31]:

$$u_x = U_0 \frac{e^{\text{Re } y/L} - 1}{e^{\text{Re}} - 1}, \quad u_y = V_0, \quad (25a)$$

$$T = T_c + (T_h - T_c) \frac{e^{\text{Pr Re } y/L} - 1}{e^{\text{Pr Re}} - 1}, \quad (25b)$$

where U_0 is the moving velocity of the top wall, $T_h (=1)$ and $T_c (=0)$ are the temperatures of the heating (top) and cooling (bottom) walls, respectively, $L (=1)$ is the width of the channel, and Re and Pr are the Reynolds number and Prandtl number, which are defined by

$$\text{Re} = \frac{V_0 L}{\nu}, \quad \text{Pr} = \frac{\nu}{\alpha}. \quad (26)$$

The analytical solution for the heat flux is

$$J_x = -\alpha \partial_x T + T u_x = U_0 \frac{e^{\text{Re } y/L} - 1}{e^{\text{Re}} - 1} \times \left[T_c + (T_h - T_c) \frac{e^{\text{Pr Re } y/L} - 1}{e^{\text{Pr Re}} - 1} \right], \quad (27a)$$

$$J_y = -\alpha \partial_y T + T u_y = V_0 \frac{e^{\text{Pr Re } T_c - T_h}}{e^{\text{Pr Re}} - 1}. \quad (27b)$$

In the simulation, boundary conditions on the bottom and top walls are treated by the nonequilibrium extrapolation approach [32,33] due to its simplicity and second-order accuracy. On the left and right sides, periodic boundary conditions are applied. The lattice spacings of the coarse and fine blocks are $\delta_{x,c} = 1/16$ and $\delta_{x,f} = \delta_{x,c}/4$, respectively. The other parameters are set as $U_0 = V_0 = 0.05$, $\text{Re} = 10$, and $\text{Pr} = 0.5$. Figure 5 gives the horizontal velocity u_x , the temperature T , and the heat flux J_x and J_y profiles along the y coordinate. It shows that the numerical results are in good agreement with the analytical solutions. The variables u_x and T are smooth across the block interface, while there is a small jump for the heat flux J_y . This error can be explained as follows. On the one hand, considering a one-dimensional steady-state problem and the special arrangement of the blocks (all the information along the boundary of the fine block remains unchangeable), no additional numerical errors are introduced by the spatial and temporal interpolations, which means the small jump cannot be due to these interpolations. On the other hand, as shown by Chai and Zhao [31], $J_x \propto \varepsilon m_3^{T,(1)}$ and

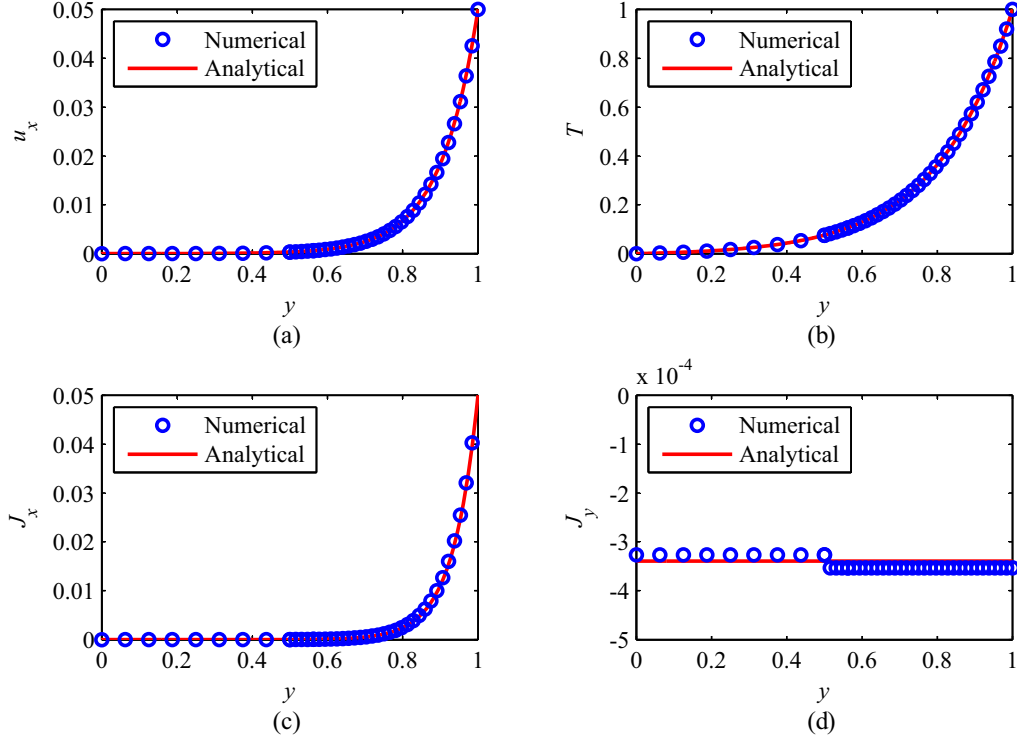


FIG. 5. (Color online) The profiles of the horizontal velocity u_x (a), the temperature T (b), and the heat flux J_x (c) and J_y (d) along the y coordinate when $U_0 = V_0 = 0.05$, $\text{Re} = 10$, and $\text{Pr} = 0.5$.

$J_y \propto \varepsilon m_5^{T,(1)}$; while when deducing the interaction approach between blocks, the following truncation $\mathbf{m}^{T,(n)} \approx \mathbf{0}$ ($\forall n \geq 2$) is adopted, which indicates that $\mathbf{m}^{T,\text{neq}} \approx \varepsilon \mathbf{m}^{T,(1)}$; this may be

the reason for the small jump. It should be pointed out that due to the tiny value of $|J_y|$ in this case, the height of the jump is apparent.

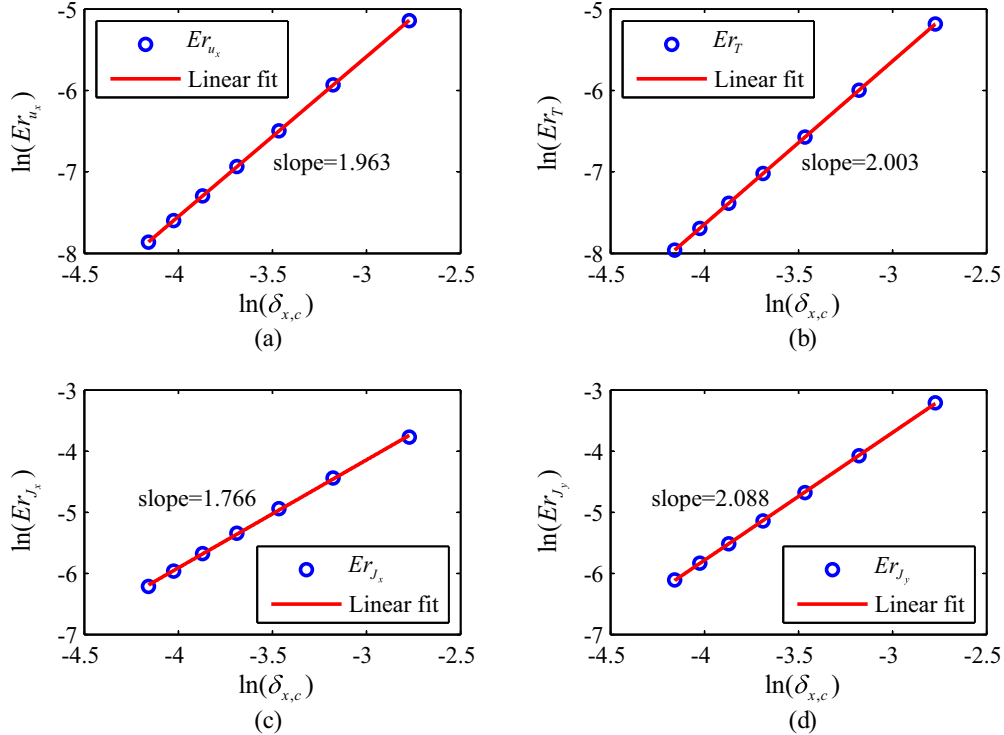


FIG. 6. (Color online) Relative errors of the horizontal velocity u_x (a), the temperature T (b), and the heat flux J_x (c) and J_y (d) versus lattice spacings $\delta_{x,c}$ when $\text{Re} = 10$ and $\text{Pr} = 0.5$.

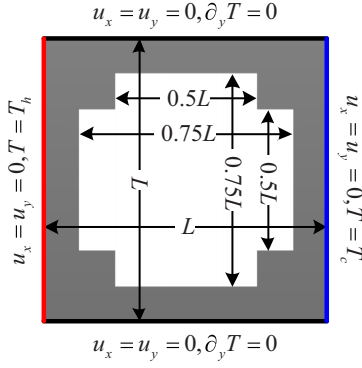


FIG. 7. (Color online) Schematic diagram of the multiblock arrangement for the natural convection in a square cavity with boundary conditions shown (the gray area is the fine block with $n = 4$, and the white area is the coarse block).

To estimate the overall convergence rate of TLBM adopting the present multiblock approach, the relative errors with different lattice spacings are calculated here. The relative errors are defined as

$$Er_\phi = \frac{\sqrt{\sum_{x,y} (\phi_{\text{numerical}} - \phi_{\text{analytical}})^2}}{\sqrt{\sum_{x,y} \phi_{\text{analytical}}^2}}. \quad (28)$$

Here, ϕ is used to represent u_x , T , J_x , and J_y ; $\phi_{\text{numerical}}$ and $\phi_{\text{analytical}}$ represent the corresponding numerical result and analytical solution of ϕ , respectively. The lattice spacing $\delta_{x,c}$ decreases from $1/16$ to $1/64$ with a corresponding $\delta_{x,f} \equiv \delta_{x,c}/4$. Considering the fact that the relaxation factors have an influence on numerical errors, they are kept unvaried. Thus fixed Re ($Re = 10$) is achieved by decreasing the velocity $V_0 (=U_0)$ when $\delta_{x,c}$ decreases ($U_0 = V_0 = 0.05$ at $\delta_{x,c} = 1/16$). Figure 6 shows the numerical results. As seen from the figures, the overall convergence rate is second order, which indicates that the adoption of the present multiblock approach does not deteriorate the convergence rate of the TLBM. What is more, from Fig. 6(d), we can infer that the convergence rate of the local scheme for computing heat flux proposed in Sec. IV is also second order, which is consistent with the convergence rate of the TLBM. Note that the overall convergence rate for computing J_x is a little lower than second order [it is 1.766, as shown by Fig. 6(c)], which is caused by the boundary condition treatment on the bottom and top walls (just as discussed by Chai and Zhao [31] and Krüger *et al.* [34,35]) rather than the adoption of the present multiblock approach.

TABLE I. Lattice spacings for different Rayleigh numbers Ra with fixed Prandtl number $Pr = 0.71$.

Ra	10^3	10^4	10^5	10^6	10^7	10^8	10^9
$\delta_{x,c}$	$1/32$	$1/64$	$1/64$	$1/128$	$1/256$	$1/512$	$1/1024$
$\delta_{x,f}$	$1/128$	$1/256$	$1/256$	$1/512$	$1/1024$	$1/2048$	$1/4096$

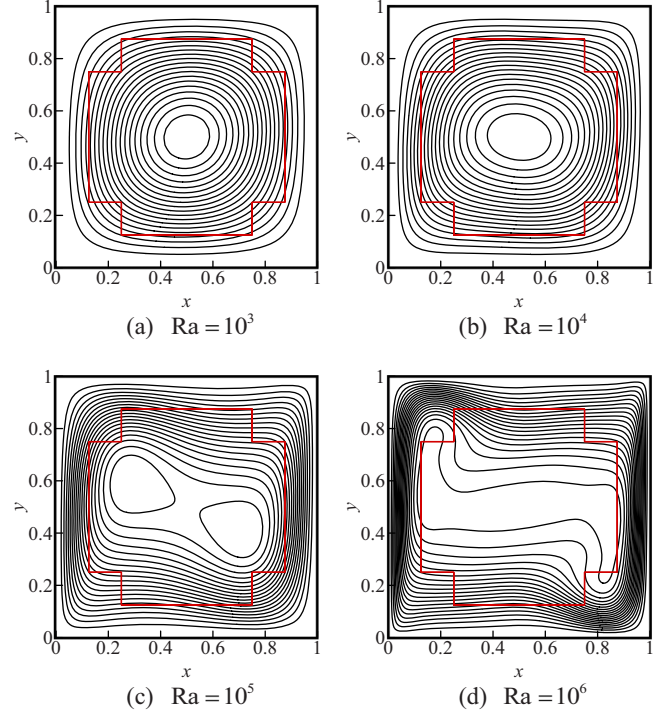


FIG. 8. (Color online) Streamlines of the natural convection in a square cavity at $Ra = 10^3$ (a), $Ra = 10^4$ (b), $Ra = 10^5$ (c), and $Ra = 10^6$ (d) (the red solid line is the inner boundary of the fine block).

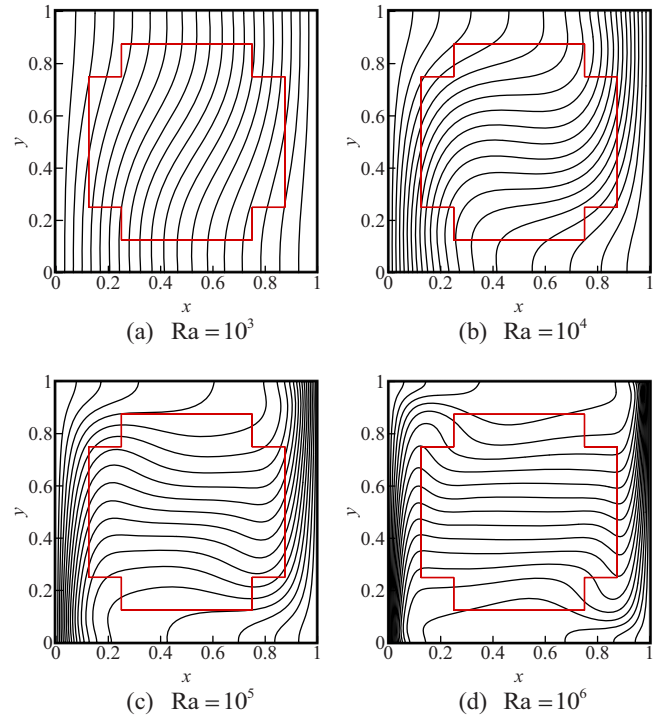


FIG. 9. (Color online) Isotherms of the natural convection in a square cavity at $Ra = 10^3$ (a), $Ra = 10^4$ (b), $Ra = 10^5$ (c), and $Ra = 10^6$ (d) (the red solid line is the inner boundary of the fine block).

TABLE II. Comparisons of the average Nusselt number (Nu) along the left side, the maximum scaled velocities ($u_{\max}$ and $v_{\max}$), and the corresponding coordinates (y_u and x_v) along the vertical and horizontal midplanes.

Ra		Nu	$u_{\max}$	$v_{\max}$	y_u	x_v
10^3	Present	1.1243×10^0	3.6565×10^0	3.7058×10^0	8.1250×10^{-1}	1.8750×10^{-1}
	Ref. [10]	1.117	3.644	3.691	0.810	0.180
	Ref. [12]	1.1195	3.643	3.6919	0.8047	0.1719
	Ref. [16]	1.1272	3.6529	3.682	0.8125	0.17183
10^4	Present	2.2497×10^0	1.6177×10^1	1.9640×10^1	8.2813×10^{-1}	1.1719×10^{-1}
	Ref. [10]	2.241	16.134	19.552	0.820	0.120
	Ref. [12]	2.2545	16.1254	19.5577	0.8203	0.1172
	Ref. [16]	2.247	16.163	19.569	0.828	0.125
10^5	Present	4.5403×10^0	3.4718×10^1	6.8718×10^1	8.5938×10^{-1}	6.6406×10^{-2}
	Ref. [10]	4.511	34.261	67.799	0.855	0.065
	Ref. [12]	4.5278	34.6033	68.0820	0.8516	0.0703
	Ref. [16]	4.5226	35.521	68.655	0.8554	0.0664
10^6	Present	8.8430×10^0	6.5034×10^1	2.2096×10^2	8.5156×10^{-1}	3.7109×10^{-2}
	Ref. [10]	8.731	63.024	215.26	0.848	0.040
	Ref. [12]	8.7746	64.9059	218.900	0.8516	0.0391
	Ref. [16]	8.805	64.186	219.866	0.8496	0.0371
10^7	Present	1.6535×10^1	1.5020×10^2	7.0107×10^2	8.7988×10^{-1}	2.1484×10^{-2}
	Ref. [16]	16.79	164.236	701.922	0.851	0.020
	Ref. [36]	16.52	148.6	699.2	0.879	0.0215
	Ref. [37]	16.92	149.61	687.20	0.900	0.021
10^8	Present	3.0219×10^1	3.2664×10^2	2.2277×10^3	9.2432×10^{-1}	1.2207×10^{-2}
	Ref. [16]	30.506	389.877	2241.374	0.937	0.0112
	Ref. [36]	30.31	322.7	2223	0.928	0.0120
	Ref. [37]	32.12	278.49	2095.23	0.930	0.009
10^9	Present	5.4628×10^1	5.8986×10^2	7.0678×10^3	9.2126×10^{-1}	6.8351×10^{-3}
	Ref. [16]	57.350	503.24	6820.07	0.966	0.0064
	Ref. [36]	—	—	—	—	—
	Ref. [37]	—	—	—	—	—

B. Natural convection in a square cavity

As a further application of the present multiblock approach and local scheme for computing heat flux, natural convection in a square cavity shown in Fig. 7 is simulated in this section. The problem is characterized by two dimensionless parameters: the Prandtl number Pr and the Rayleigh number Ra, which are defined as

$$\text{Pr} = \frac{\nu}{\alpha}, \quad \text{Ra} = \frac{|\mathbf{g}| \beta (T_h - T_c) L^3}{\nu \alpha}, \quad (29)$$

where $T_h (=1)$ and $T_c (=0)$ are the temperatures of the heating (left) and cooling (right) walls of the cavity, respectively, and $L (=1)$ is the length of side for the square cavity. Considering that the velocity and temperature boundary layers develop along the heating and cooling walls, as well as the direction of fluid motion turns rapidly around the four corners, the fine block with $n = 4$ is arranged in the vicinity of all sides and corners of the square cavity, as illustrated by Fig. 7.

In the simulation, both the velocity and temperature boundary conditions are treated by the nonequilibrium extrapolation approach [32,33]. The gravity acceleration is $|\mathbf{g}| = 1.0$, and the characteristic velocity $U_c = \sqrt{|\mathbf{g}| \beta (T_h - T_c) L}$ is fixed at 0.2 to accelerate the convergence as well as to satisfy the condition

of the small Mach number. The other parameters are the Prandtl number Pr, which is fixed at 0.71, and the Rayleigh number Ra, which ranges from 10^3 up to 10^9 . As shown in Table I, the lattice spacing adopted for the simulation decreases as the Rayleigh number increases to ensure the numerical stability and accuracy.

Figures 8 and 9 show the streamlines and isotherms for low Rayleigh numbers (i.e., $\text{Ra} \leq 10^6$), respectively. In this region, the convection is laminar flow and a steady convergent result can be obtained easily. With regard to the flow pattern, it can be found from Fig. 8 that a single vortex appears in the center of the cavity at $\text{Ra} = 10^3$; as Ra increases, the vortex tends to become elliptic ($\text{Ra} = 10^4$) and breaks up into two vortices at $\text{Ra} = 10^5$; when $\text{Ra} = 10^6$, the two vortices move to the heating and cooling walls oppositely and another vortex appears in the center of the cavity. With regard to the heat transfer mechanism, Fig. 9 indicates that when Ra is relatively small (i.e., $\text{Ra} = 10^3$), conduction is the dominating mechanism, which is evident from the nearly vertical isotherms; as Ra increases, the role of convection grows progressively and the isotherms tend to become horizontal in the center of the cavity. In Figs. 8 and 9, it can be seen that both the streamlines and isotherms exhibit a smooth transition across the block interface. In order to further

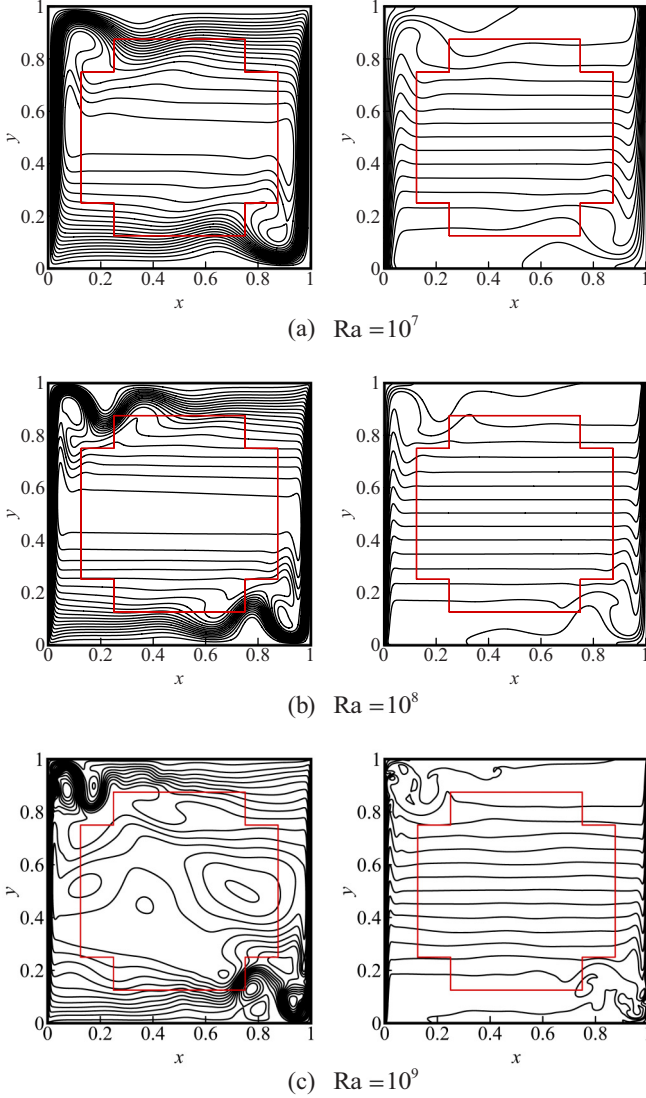


FIG. 10. (Color online) Streamlines (left) and isothermals (right) of the natural convection in a square cavity at $Ra = 10^7$ (a), $Ra = 10^8$ (b), and $Ra = 10^9$ (c) (the red solid line is the inner boundary of the fine block).

validate the present multiblock approach, as well as the present local scheme for computing heat flux, the average Nusselt number (Nu) along the left side, the maximum scaled velocities ($u_{\max}$ and $v_{\max}$, scaled by α/L), and the corresponding coordinates (y_u and x_v) along the vertical and horizontal midplanes are given in Table II. As shown, the present results agree well with the data from previous studies.

Figure 10 shows the streamlines and isothermals for high Rayleigh numbers (i.e., $Ra \geq 10^7$). In this region, the convection tends to become turbulent flow and a convergent result is not easy to obtain when uniform grid is used because the computation load may increase rapidly. However, by adopting the multiblock approach, the computation load can be decreased significantly without losing the numerical accuracy. From Fig. 10, we can find that the bulk of the fluid motion takes place in the regions close to the heating and

cooling walls and secondary vortices develop at the upper-left and bottom-right corners. As Ra increases, the temperature gradient along the heating and cooling walls becomes steeper, which indicates a stronger heat transfer. All these observed flow and heat transfer characteristics conform well to the results reported in previous works [16,36]. Note that when $Ra = 10^9$, the convection becomes unsteady, and Fig. 10(c) shows the instantaneous streamlines and isothermals at some time point. This is consistent with previous works which show that the natural convection becomes unsteady for Ra higher than 1.9×10^8 [38,39]. Again, it can be seen from Fig. 10 that the streamlines and isothermals are smooth across the block interface. To further validate the present multiblock approach and local scheme for computing heat flux, Nu , $u_{\max}$, $v_{\max}$, y_u , and x_v (the time-averaged values for $Ra = 10^9$) are also given in Table II. Good consistency with the data from previous studies can be seen again.

VI. CONCLUSION

In this paper, we have proposed a multiblock approach for the passive scalar thermal lattice Boltzmann method with multiple-relaxation-time collision scheme, and have considered an external force term in LBEs. The interaction approach between blocks is developed based on the continuity of the first three order (ε^0 , ε^1 , and ε^2) expansion equations in the Chapman-Enskog analysis, which is more fundamental and shows that all the nonequilibrium parts of the nonconserved moments should be rescaled and the nonequilibrium parts of the conserved moments can be calculated directly. What is more, a local scheme for computing heat flux has also been proposed based on the pseudoparticles, with no need to calculate temperature gradient based on the finite-difference scheme. Numerical tests have been carried out to validate the proposed multiblock approach and local scheme for computing heat flux. Smooth results can be obtained except that there is a small jump for heat flux across the block interface which may be caused by the truncation of second order (ε^2) when rescaling the nonequilibrium parts of the nonconserved moments. Moreover, the numerical results show that the adoption of the present multiblock approach does not deteriorate the convergence rate of the TLBM, and the present local scheme for computing heat flux also has second-order convergence rate. Further applications show that natural convection in a square cavity with severe condition ($Ra \geq 10^7$) can be simulated effectively with the multiblock approach due to the decrease of computation load.

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